

Lesson 6 Quick Reference — Comments, Style & Helpers

Syntax only. For the *why*, see the Lesson 6 reference sheet.

Comments & docstrings

```
# one line only - Python has no /* block */

"""
Docstring: what this FILE does. One, at the top.
"""
```

	Use it for
"""docstring"""	what the whole file does
# comment	why a line exists

Say **why**, not what.

```
WIDTH = 34 # fits a phone screen — not # set WIDTH to 34
```

Every submitted file needs:

```
# Your Name | Assignment | Date
```

PEP 8 — the style rules

1. `snake_case` variables, `ALL_CAPS` constants
2. 4 spaces to indent — **never tabs**
3. Spaces around operators: `x = 1 + 2`
4. One statement per line

5. Lines under 79 characters

```
totalCost=price*qty+fee          # runs, loses marks
total_cost = price * qty + fee    # PEP 8
```

Helpers

```
len("PYTHON")          # 6      characters in a string
round(19.994, 2)        # 19.99 round to 2 decimal places
round(9.8)              # 10     round to nearest whole
int(9.8)                # 9      chop off the decimal
```

int() chops · round() rounds

round(3.14159, 2) → 3.14

Shorter assignment

```
total += 10              # same as total = total + 10
count -= 5               # count = count - 5
n *= 2                  # n = n * 2
x /= 4                  # x = x / 4
y //= 2                 # y = y // 2
z %= 2                  # z = z % 2
p **= 2                 # p = p ** 2
```

Each line uses the value from the line before it.

n = 4; n += 3; n *= 2 → 14

Don't repeat yourself

```
TICKET_PRICE = 12.50      # a number used twice → name it
```

Error → cause

Error	It means
<code>IndentationError</code>	mixed tabs and spaces
Python Formatters underline a name	<code>camelCase</code> — use <code>snake_case</code>
<code>SyntaxError</code> on <code>/* */</code>	Python has no block comment
money off by a penny	used <code>int()</code> where <code>round()</code> belongs

Before you push

header comment · docstring · `snake_case` · spaces around operators · repeated numbers named ·
`total = total + x` replaced with `+=` · comments say why · **pushed**